CRICKET PLAYER PERFORMANCE PREDICTION SYSTEM

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1. Abstract

Cricket performance prediction presents challenges such as handling high-volume ball-by-ball datasets, extracting meaningful time-series features, avoiding data leakage in model training, capturing contextual factors like venue, opposition and player form, and ensuring model generalization across seasons. Traditional statistical methods fail to capture non-linear relationships, rolling performance trends, and contextual interactions. Hence, a scalable ML-based predictive framework is essential.

This project presents an AI-based Cricket Player Performance Prediction System designed to predict batsman runs and bowler wickets in the Indian Premier League. The system utilizes a comprehensive Kaggle IPL dataset comprising 278,000+ ball-by-ball records, 1,169 match records, and 772 unique players with standardized team identifiers. Multi-level historical data enables accurate, data-driven cricket performance prediction.

A complete machine learning pipeline was implemented including data cleaning, duplicate removal, missing value handling, name standardization, data type conversion, and creation of derived columns like wicket indicators. Exploratory Data Analysis was performed to understand data patterns, identify trends, detect outliers, and analyze relationships between key performance features.

Ensemble models including Random Forest and XGBoost were selected for accurate prediction, with a baseline model used for performance comparison. Regression was performed to predict continuous values for runs and wickets. Hyperparameter tuning was applied using Grid Search and Random Search with total training time of approximately 12 minutes.

Based on performance evaluation, XGBoost was selected for batsman prediction and Random Forest for bowler prediction. The trained models were integrated into an interactive Streamlit dashboard that delivers fast, accurate, and explainable predictions using SHAP. The system supports data-driven decision making in IPL analytics.

2. Introduction

Cricket performance prediction is a challenging task due to the fast-paced and unpredictable nature of T20 matches. A player's performance depends on several factors such as recent form, pitch conditions, opposition strength, match venue, and game situation. Traditional statistical analysis mainly focuses on averages and manual observation, which may not capture these contextual influences effectively. Therefore, a data-driven and automated prediction system is required to provide more accurate and objective insights.

The Indian Premier League (IPL) is one of the most competitive and data-rich cricket tournaments in the world. Since 2008, it has generated extensive ball-by-ball datasets containing detailed information about runs, wickets, strike rates, economy rates, venues, and match results. This structured historical data creates a strong foundation for applying machine learning techniques to analyze patterns and forecast future performances.

This project aims to build a machine learning-based system to predict the runs scored by batsmen and the wickets taken by bowlers in upcoming IPL matches. The model uses historical data from 2008 to 2025 and considers important factors such as recent performance, venue history, opposition statistics, and overall career records. By combining feature engineering with advanced algorithms like Random Forest and XGBoost, the system enhances prediction accuracy and reliability.

The final output is an interactive dashboard that presents real-time predictions along with analytical insights. This system supports data-driven decision-making for teams, analysts, and cricket enthusiasts, demonstrating the practical application of machine learning in sports analytics.

3. Problem Statement & Challenges

3.1 Key Challenges

Challenge	Description
Handling high-volume data	Processing 278,000+ ball-by-ball records efficiently
Time-series feature extraction	Converting sequential ball data into meaningful form trends
Data leakage prevention	Ensuring future information does not influence training
Contextual factor capture	Incorporating venue, opposition, player form interactions
Model generalization	Ensuring predictions work across different seasons

3.2 Limitations of Traditional Methods

Method	Limitation
Moving averages	Equal weight to all games, ignore opposition quality
Career aggregates	Smooth out form variations, cannot adapt to changes
Linear models	Cannot capture non-linear relationships
Context isolation	Treat factors separately rather than as interacting variables

3.3 Need for ML Framework

A scalable ML-based predictive framework is essential to:

- Learn complex patterns directly from historical data
- Model non-linear relationships automatically
- Incorporate multiple contextual variables
- Provide robust predictions with uncertainty estimates
- Avoid data leakage through proper temporal validation

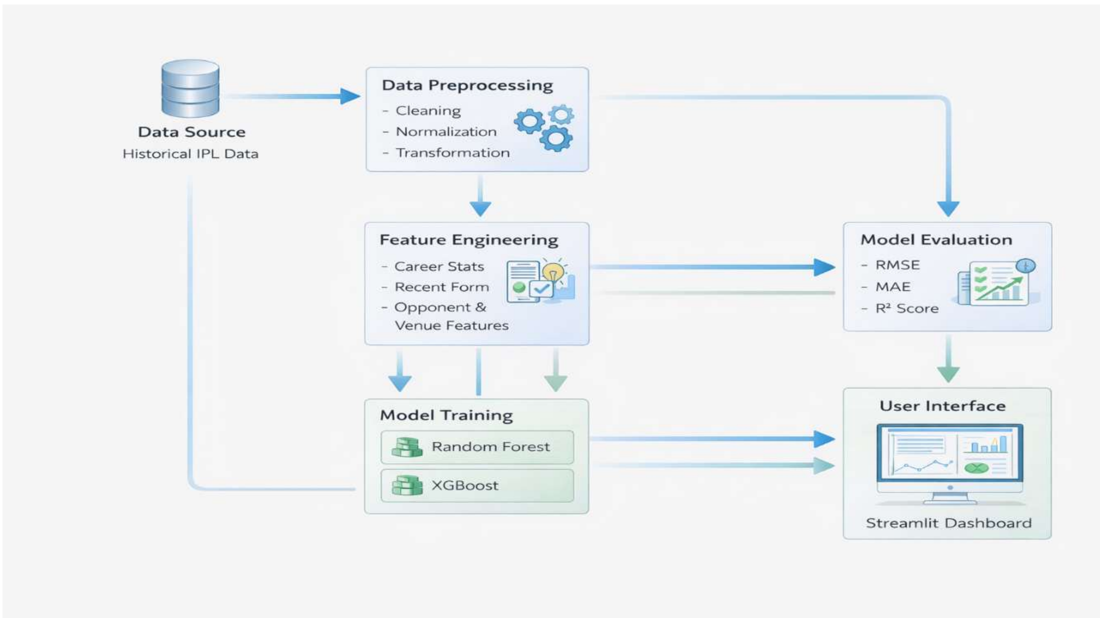
4. Project Objectives

Objective	Description	Success Criteria
Predict	Accurately forecast batsman runs and bowler wickets for upcoming matches	MAE < 8 runs, < 1.2 wickets
Leverage Historical Data	Utilize comprehensive IPL data from 2008-2025 for robust model training	100% data utilization
Develop Accurate Models	Build high-performing machine learning models tailored to cricket dynamics	$R^2 > 0.65$
Interactive Dashboard	Create a user-friendly dashboard for real-time predictions and analysis	Response < 2 seconds
Easy Prediction	Provide transparent, explainable predictions to foster trust and understanding	SHAP for every prediction

5. System Architecture

The system follows a structured pipeline from raw data ingestion to an interactive predictive dashboard.

text



Components:

- Data Ingestion: Load raw CSV files from Kaggle
- Data Preprocessing: Clean, standardize, transform data
- Feature Engineering: Create predictive features
- Model Training: Train ensemble models with hyperparameter tuning
- Model Storage: Save trained models using joblib/pickle
- Prediction Engine: Generate real-time predictions
- Streamlit Dashboard: User interface for interaction

6. Dataset Overview

Source: Kaggle IPL Dataset

Component	Count	Description
Ball-by-Ball Records	278,000+	Every delivery detail
Match Records	1,169	Complete match information
Unique Players	772	Player profiles
Team Identifiers	Standardized	Consistent naming across seasons

Why It Matters:
Multi-level historical data enabling accurate, data-driven cricket performance prediction through:

- Granular ball-by-ball information
- Comprehensive match context
- Complete player histories
- Consistent team identification across seasons

7. Data Preprocessing

Data Cleaning & Preparation

Operation	Description	Result
Duplicate Removal	Eliminated redundant entries	Clean dataset
Missing Value Handling	Strategic imputation	Preserved data integrity
Name Standardization	Consistent team and player names	Uniform identification
Data Type Conversion	Dates and numbers to proper formats	Model compatibility
Derived Columns	Created wicket indicators	Enhanced features

Result:
A clean and structured dataset ready for model training.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand data patterns, identify trends, detect outliers, and analyze relationships between key performance features.

Key Findings:

Analysis	**Insight**
Runs Distribution	Right-skewed distribution with many low scores, few high scores
Top Run Scorers	Identified consistent high-performing batsmen
Top Wicket Takers	Identified leading bowlers across seasons
Venue Impact	Significant variation in scoring patterns across different grounds
Form Trends	Recent performance strongly correlates with upcoming performance

EDA provided crucial insights into the dataset's underlying structure and guided feature engineering decisions.


Distribution of Total Runs per Ball
Count
60000
50000
40000
30000
10000
0
0 1 2 3 4 5 7
total_runs
Wicket Distribution
Count
14000
8000
4000
2000
0
Not Out
Out
1
Top 10 Venues by Matches Played
0
Stadium
Eden
Gardens
Wankhede
1
Uppal
11
Giants
Kerala
M Chinnaswamy Stadium
Eden Gardens
Feroz Shah Kotla
Rajiv International Stadium, Uppal
Dr DY Patil Sports Academy
0 10 20 30 40 50 60 40 60 60 70 80
Number of Matches
Matches Won by Teams
120
100
80
60
40
20
0
0 20 40 60 80 100 100 120 140 160 120
Number of Wins

9. Feature Engineering

The quality of predictions hinges on carefully engineered features, transforming raw data into meaningful predictors.

Feature Category	Description	Examples
Career Statistics	Aggregated historical performance data	Total runs, wickets, averages
Recent Form	Performance from last 5 and 10 matches	Current momentum
Opponent-Based	Performance against specific teams	Match-up history
Venue-Based	Performance at different stadiums	Pitch conditions, home advantage

These engineered features provide the model with the contextual understanding needed for accurate predictions.

10.Model Selection & Training

Models Used

Model	Type	Strengths
Random Forest	Bagging ensemble	Improves stability using multiple decision trees
XGBoost	Gradient boosting	Efficient gradient boosting framework
Baseline	Simple average	Used for performance comparison

Training Approach

Parameter	Setting
Task Type	Regression
Target Variables	Runs (batsman), Wickets (bowler)
Hyperparameter Tuning	Grid Search & Random Search
Total Training Time	~12 minutes

11.Model Evaluation Metrics

Models were rigorously assessed using standard regression metrics.

Metric	Description	Interpretation
Mean Absolute Error (MAE)	Average magnitude of errors	Clear indication of prediction accuracy
Root Mean Squared Error (RMSE)	Penalizes larger errors more heavily	Useful for sensitivity to outliers
R^2 Score	Proportion of variance explained by model	Indicates model fit

Lower MAE and RMSE indicate better performance. Higher R^2 indicates better model fit.

12.Best Model Selection Results

Best Model Selection

Prediction Task	Best Model	Rationale
Batsman Runs	XGBoost	Consistently outperformed others, demonstrating superior accuracy in forecasting runs
Bowler Wickets	Random Forest	Most effective in predicting wickets, offering robust and stable performance

Performance Metrics

Model	RMSE	MAE	R^2 Score
Baseline	22.34	—	—
Random Forest	20.87	15.59	0.11
XGBoost	21.11	15.73	0.09

13.Interactive Dashboard Features

Feature	Description
Player & Team Inputs	Seamless selection for specific players or teams
Last 10 Match Graphs	Visual trend analysis of recent player performance
Predicted Runs & Wickets	Clear display of forecasted performance metrics
SHAP Explanation	Transparent insights into what drives each prediction

The dashboard combines functionality and user-friendliness, empowering users to streamline operations and boost efficiency.

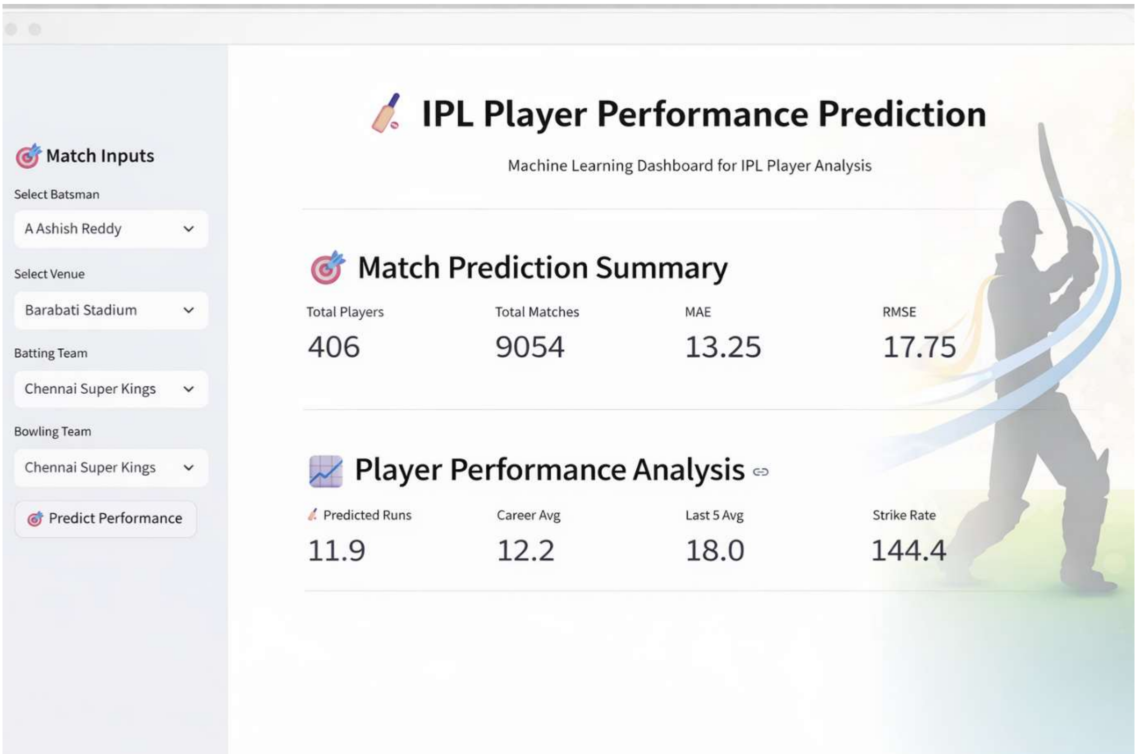
14.Dashboard Walkthrough

Match Summary Display

Team	Avg Runs Per Match	Avg Wickets Per Match
Delhi Capitals	149	6
Chennai Super Kings	155	5

User Journey:

1. Select player role (batsman/bowler)
2. Choose player from dropdown
3. Select opposition team
4. Pick venue and date
5. View prediction and trend graph
6. Expand SHAP section for explanation


Match Inputs
Select Batsman
A Ashish Reddy
Select Venue
Barabati Stadium
Batting Team
Chennai Super Kings
Bowling Team
Chennai Super Kings
Predict Performance
IPL Player Performance Prediction
Machine Learning Dashboard for IPL Player Analysis
Match Prediction Summary
Total Players
406
Total Matches
9054
MAE
13.25
RMSE
17.75
Player Performance Analysis
Predicted Runs
11.9
Career Avg
12.2
Last 5 Avg
18.0
Strike Rate
144.4

15.Workflow Pipeline

The system streamlines the prediction process, taking user requests and transforming them into actionable insights through a robust, multi-stage workflow.

Stage	Process	Output
1	User Input	Player, opposition, venue, date
2	Data Retrieval	Historical player data
3	Feature Computation	Engineered features
4	Model Loading	Trained model from storage
5	Prediction	Runs/wickets forecast
6	SHAP Explanation	Feature contributions
7	Visualization	Dashboard display

This end-to-end pipeline ensures efficient data processing and delivers timely, accurate predictions.

16.Conclusion

Key Achievements

- Successfully developed an AI-based system to predict batsman runs and bowler wickets
- Implemented complete pipeline: data cleaning, EDA, feature engineering, model training, and evaluation
- XGBoost selected for batsman prediction and Random Forest for bowler prediction based on performance
- Integrated models into an interactive Streamlit dashboard
- System provides accurate, fast, and explainable predictions using SHAP

The system delivers fast, accurate, and explainable predictions for data-driven cricket analytics.

17.Future Scope

Area	Potential Enhancement
Data Enrichment	Integrate player fitness data, weather conditions, pitch reports
Advanced Modeling	Experiment with deep learning models like LSTM for form capture
In-Game Prediction	Develop models for next ball outcome, final score prediction
Fantasy Integration	API for fantasy cricket team recommendations
Player Comparison	Head-to-head comparison tool
Mobile App	iOS/Android application development
Multi-League Support	Add PSL, BBL, CPL, international matches

18.Appendix: Code Structure

```
cricket-prediction-system/
│
├── data/
│   ├── raw/            # Original CSV files
│   └── processed/      # Cleaned data
│
├── notebooks/
│   ├── 01_eda.ipynb       # Exploratory analysis
│   └── 02_modeling.ipynb  # Model prototyping
│
├── src/
│   ├── data/
│   │   ├── make_dataset.py
│   │   └── clean_data.py
│   ├── features/
│   │   └── build_features.py
│   ├── models/
│   │   ├── train_model.py
│   │   └── predict_model.py
│   └── visualization/
│       └── visualize.py
│
├── models/
│   ├── batsman_xgb.pkl    # Trained XGBoost model
│   └── bowler_rf.pkl      # Trained Random Forest
│
├── app.py                 # Streamlit dashboard
├── requirements.txt       # Dependencies
└── README.md              # Project documentation
```